

Used Car Price Prediction

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Business Problem

The used car price is affected by several factors such as vehicle age, transmission, mileage, fuel type, seller type etc. Determining a suitable used car market value can be difficult because these variables interact with one another and market conditions constantly change. Buyers want to avoid paying more than a vehicle is worth, while sellers and dealerships aim to price vehicles competitively. The objective of this project is to develop machine learning models that predict used car selling prices and identify the factors that have the greatest influence on vehicle value. Accurate price prediction can support better decision making for buyers, sellers, and the dealerships.

Research Questions

1. What are the vehicle key features that have high impact on sale prices?
2. How does vehicle's age and mileage affect the price of used vehicles?
3. Which predictive model performs best in price prediction?

Background/History

The used vehicle market is one of the fastest-growing segments of the automotive industry. Online marketplaces have expanded the availability of auto information and making data-driven pricing more important than ever. Traditionally, vehicle prices have been estimated using experience, market comparisons, manual inspection etc. However machine learning helps an organization to analyze historical sales data and determine complex relationships between vehicle characteristics and selling prices. Predictive models could improve pricing consistency, reduce human bias, and support for better purchasing and selling decisions.

Data Explanation

This project uses the Vehicle Dataset by CarDekho sourced from open source Kaggle. The dataset contains about 4,340 used vehicle records and includes variables such as vehicle name, manufacturing year, selling price, kilometers driven, fuel type, seller type, transmission type, and ownership history. During data preparation, the dataset was examined for missing values, inconsistent records, and incorrect data types. An additional feature `car_age` was created from the manufacturing year to improve model performance. The prepared dataset was then used for exploratory data analysis and predictive modeling.

Methods

The analysis was performed using Python. The initial data preparation includes, checking for missing values, validating data types, and creating the `car_age` feature from the manufacturing year. Exploratory Data Analysis (EDA) is conducted to understand the distribution of selling prices and examine relationships between important variables such as vehicle age, mileage, fuel type, and transmission. After pre-processing, the dataset was divided into two parts, training and testing sets. Predictive models - Linear Regression and Random Forest Regression is created and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). These evaluation metrics were selected to compare prediction accuracy and overall model performance.

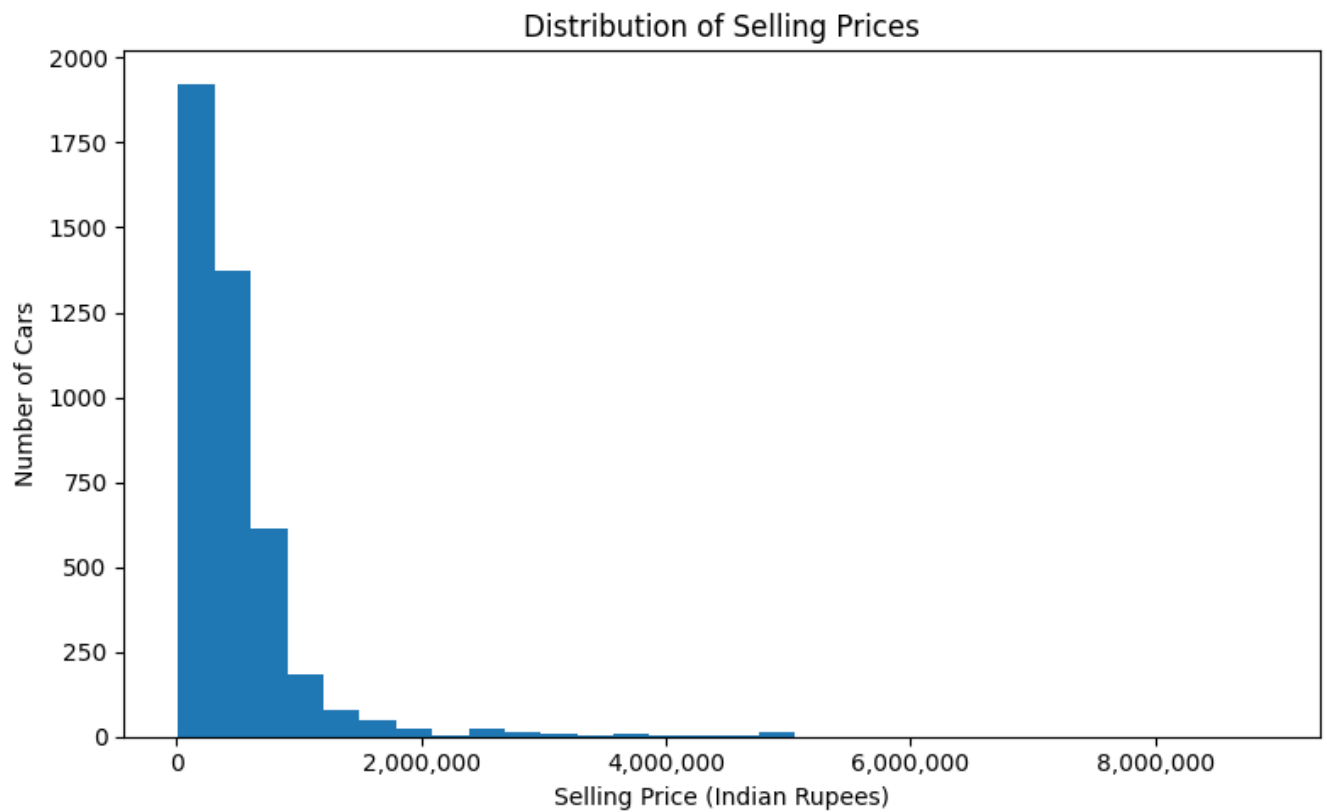
Analysis

The dataset was analyzed utilizing 5 different visualizations for better understanding to find out the relationships between vehicle characteristics and selling price. The histogram of selling prices shows that most vehicles are concentrated in the lower price range, while relatively few vehicles have very high selling prices. This right-skewed distribution suggests that lower-

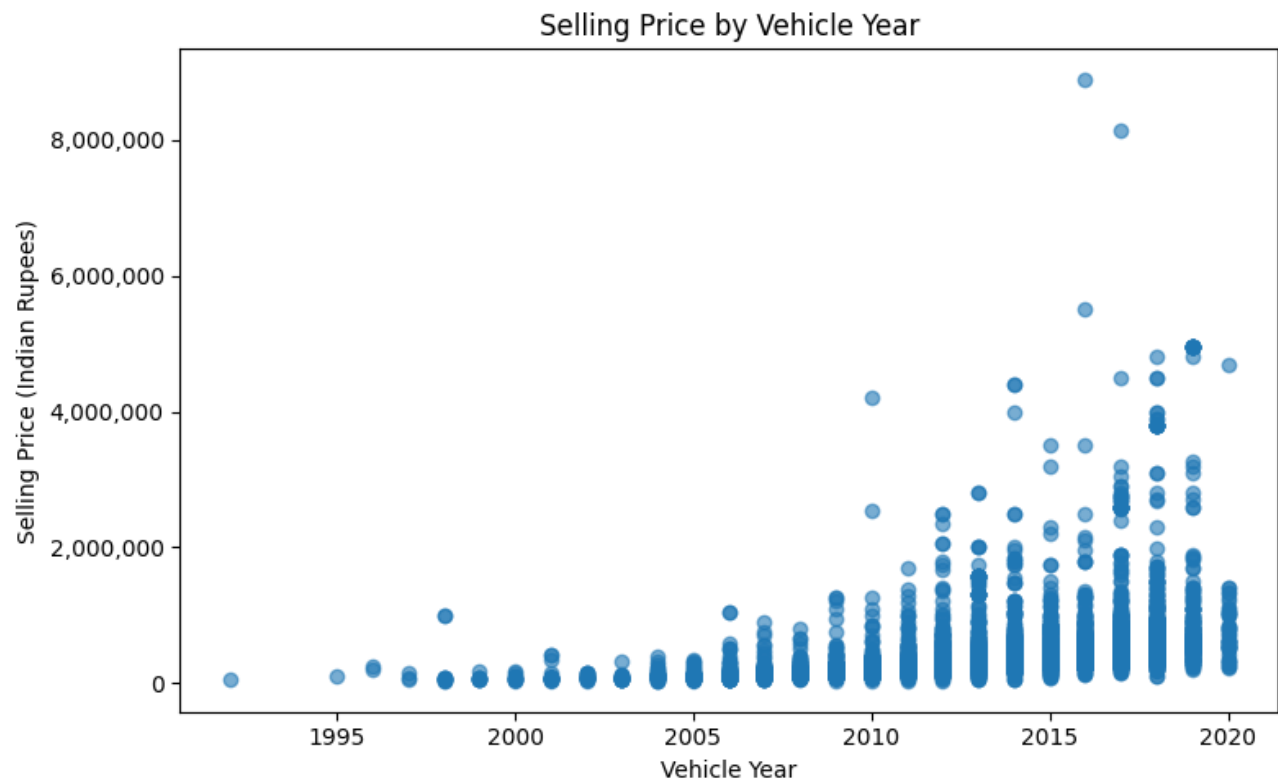
priced used cars dominate the dataset. The CarDekho dataset represents the Indian used-car market so the findings presented in this analysis may not fully reflect vehicle trends in other countries, such as the United States.

Visualization

Figure 1. Distribution of Selling Prices

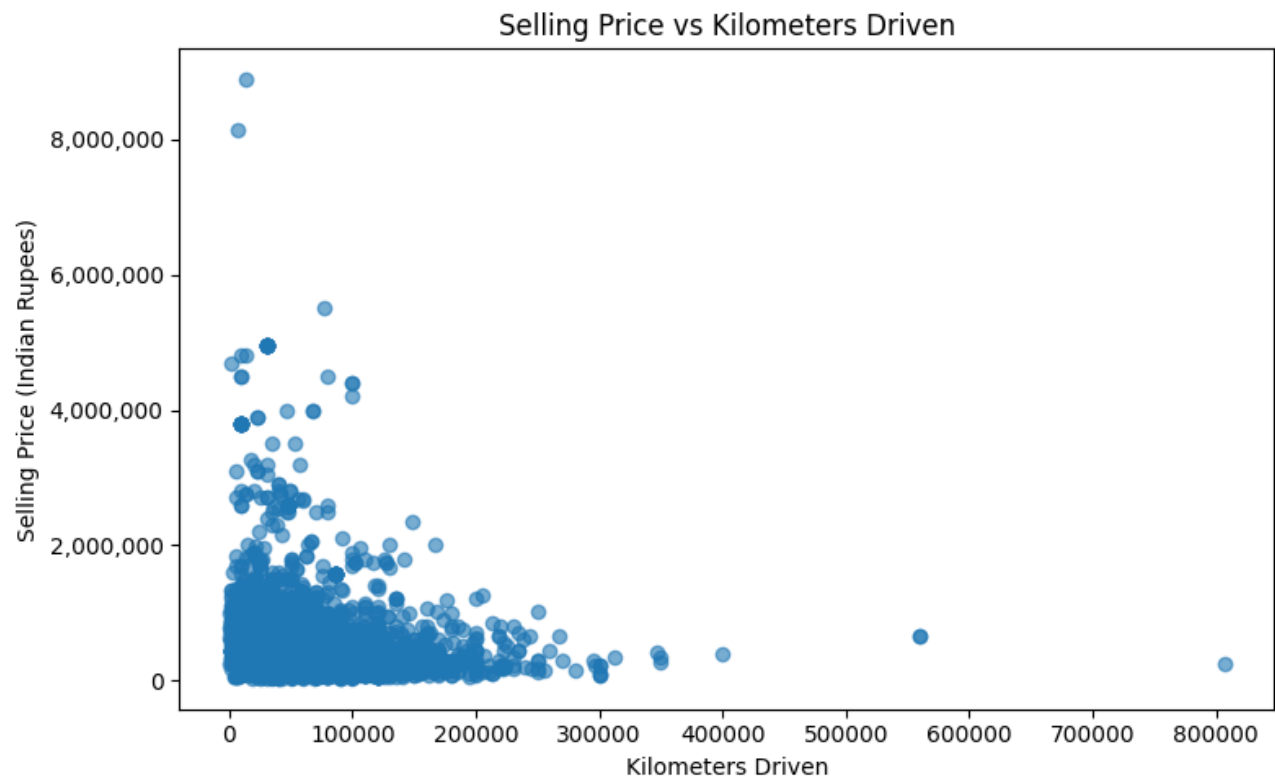


The histogram shows that most of the vehicles are concentrated in the lower selling price ranges, while few vehicles are sold at very high prices. This right-skewed distribution indicates that lower priced used cars make up the majority of the dataset.

Figure 2. Selling Price by Vehicle Year

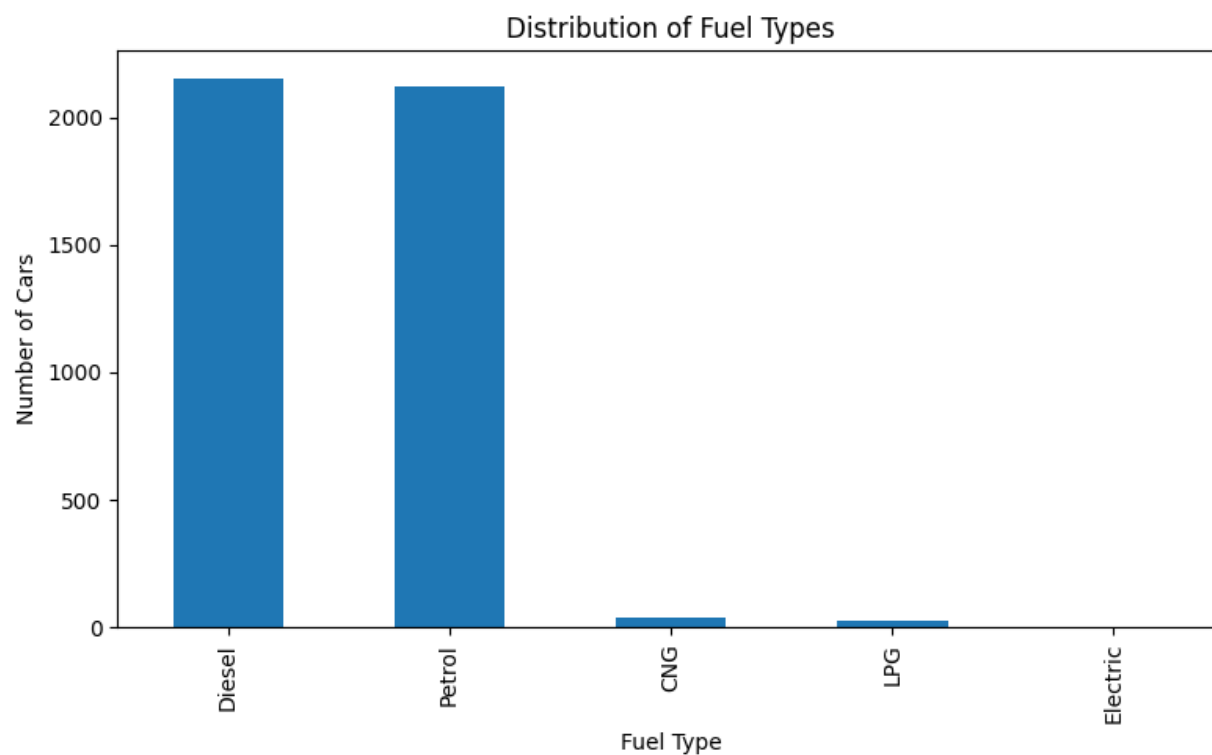
It shows that the newer vehicles generally have higher selling prices than older vehicles. Although prices vary within each model year, there is an overall upward trend, indicating that vehicle age is an important factor influencing selling price.

Figure 3. Selling Price vs. Kilometers Driven

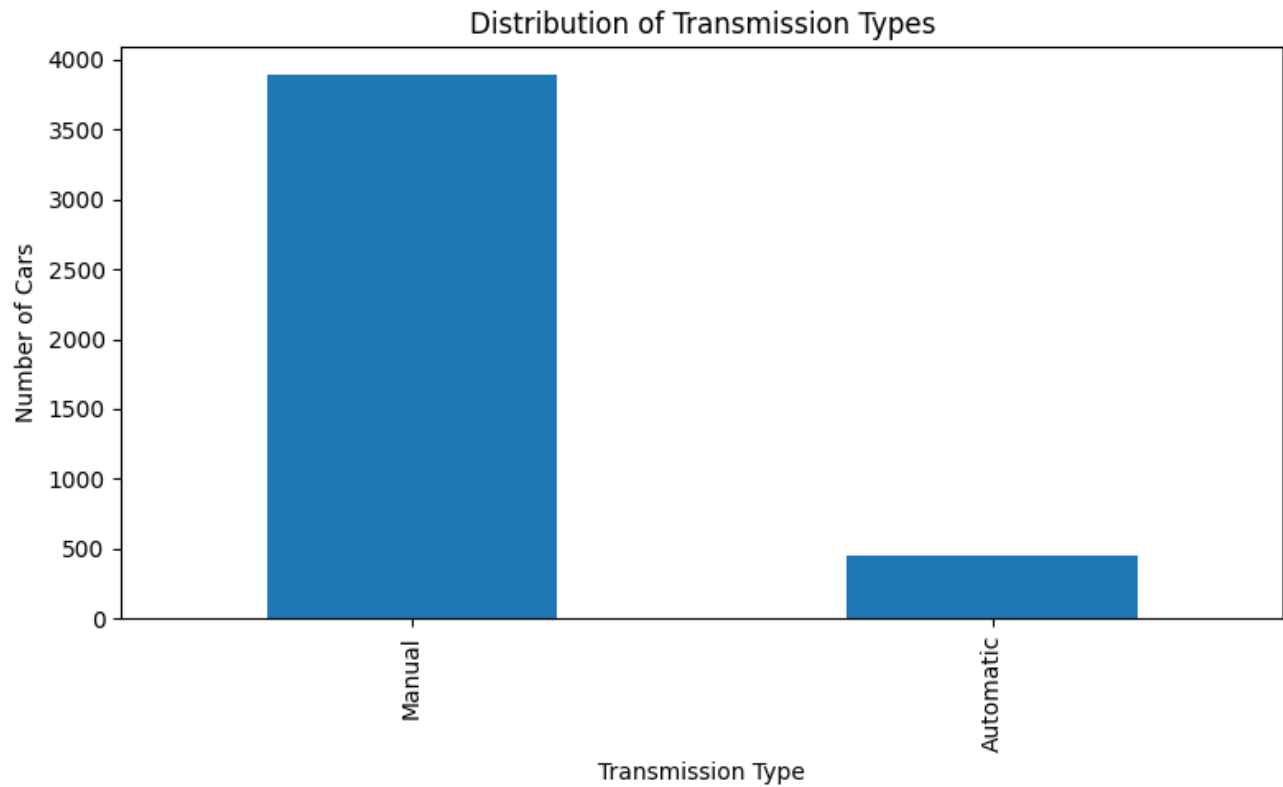


It indicates the higher mileage generally have lower selling prices. The overall trend indicates a negative relationship between kilometers driven and selling price.

Figure 4. Distribution of Fuel Types



The bar chart shows that the diesel and petrol vehicles make up the majority of the dataset, while CNG, LPG, and electric vehicles are relatively rare.

Figure 5. Distribution of Transmission Types

The chart shows that the manual transmission vehicles make up the majority of the dataset, while automatic transmission vehicles are much less common.

To evaluate prediction performance, Linear Regression and Random Forest Regression models were created and compared by using Mean Absolute Error, Root Mean Squared Error, and the coefficient of determination (R^2). The results is summarized in the Table 1.

Table 1. Model Performance Comparison

Model	MAE	RMSE	R^2
Linear Regression	221,706.37	426,786.69	0.403
Random Forest Regression	168,075.10	390,833.86	0.499

The Random Forest Regression model outperformed Linear Regression model for all three performance measures. In the Random Forest Regression model, the values of MAE and RMSE were smaller which is an indication of less prediction error, whereas the value of R^2 was larger, implying a better fitting of the model to the data.

Conclusion

This project revealed that machine learning can be used to predict used car prices using historical vehicle data. Exploratory Data Analysis showed that vehicle age, kilometers driven, fuel type, transmission type, and ownership history all contribute to vehicle value. Among the two models evaluated, Random Forest Regression produced better prediction accuracy than Linear Regression, making it the preferred model for this dataset.

Although the model produced encouraging results, prediction accuracy can improve by incorporating additional variables such as accident history, maintenance records, geographic location, vehicle condition and service history. Because this study used the used car dataset of India, the results may not fully represent used-car markets in other countries, such as the United States.

Overall, this study reveals that machine learning provides a practical approach for supporting pricing decisions and can assist buyers, sellers, and dealerships in making more informed decisions.

Assumptions

It is assumed that the data provided is an accurate representation of used cars and that recorded values are reliable. Furthermore, it is assumed that information from historical prices patterns will be helpful for making predictions.

Limitations

The dataset was collected from one source only, and some variables like accident, maintenance, location, and car conditions have not been considered in this analysis. This missing variables may affect prediction accuracy.

Challenges

Some challenges are handling outliers, picking the effective variables, and making a trade-off between model complexity and interpretability.

Future Uses/Additional Applications

The model can be used for making pricing decisions at car dealerships, online valuation tools, inventory management, insurance estimations, and consumer purchasing decisions.

Recommendations

Future studies should involve further feature engineering, hyperparameter tuning and testing of other sophisticated machine learning models such as XGBoost and Gradient Boosting. The data can be expanded by considering cars from different countries and adding some other features about the vehicles for better predictions.

Implementation Plan

1. Complete data preparation.
2. Improve the predictive models.
3. Compare model performance.
4. Interpret the findings.
5. Finalize the report and presentation.

Ethical Assessment

The dataset used for this project does not have any information that is sensitive to the individuals. Predictive models can only be used as decision support tools but not as a replacement for human judgement. There may be historical bias present in datasets that will affect the prediction generated by the model. In addition, predictions should not be directly applied to other markets without further validation because the dataset represents the Indian used-car market only.

References

Kaggle. (n.d.). *Vehicle Dataset from CarDekho*.

<https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho>

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.